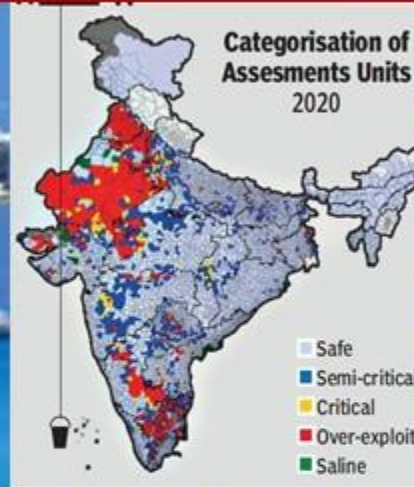


INDIAN ECONOMY

LEVEL
II



LECTURES

- REFORMS IN THE SPACE SECTOR
- LIBERALISED DRONE RULES, 2021
- KEN-BETWA RIVER LINK
- SVAMITVA

MODULE - 50



REFORMS IN THE SPACE SECTOR TOWARDS MORE AND MORE PRIVATISATION!

- Since its inception in the 1960s, Indian Space Program has grown tremendously, administered by the Department of Space and primarily executed by its R&D arm, the Indian Space Research Organization (ISRO).
- Capabilities have been developed in the space sector across all domains. These include
 - ✓ Indigenous space transportation systems, like PSLV and GSLV.
 - ✓ Space assets comprising of a fleet of satellites, catering to the needs of earth observation, satellite communication, meteorology, space science and navigations.
 - ✓ Ground infrastructure and a host of operational programs related to the applications of space technology to the common uses of the society.

- Across the globe, the space activities are in a state of transition towards private sector.
- From being primarily a Government driven activity, the sector has been witnessing increasing participation of private sector not only in the traditional vendor role, but also in taking up end to end space activities.

Keeping this in mind, Government undertook reforms in the space sector in 2020.



REFORMS IN THE SPACE SECTOR

START-UPS IN SPACE SECTOR ARE INCREASING EXPONENTIALLY!

Year	No of start-ups
2012	1
2013	1
2014	1
2015	3
2016	1
2017	8
2018	7
2019	11
2020	21
2021	47
Total	101

NEW SPACE INDIA LIMITED (NSIL)

- New space India Limited (NSIL), the Public Sector Undertaking, was launched to own the operational launch vehicles and the space assets of ISRO.
- The present supply-based model was changed to demand-driven model, where in, NSIL shall act as an aggregator of user requirements and obtain commitments.



REFORMS IN THE SPACE SECTOR

INDIAN NATIONAL SPACE PROMOTION AND AUTHORIZATION CENTER (IN-SPACE)



- Independent Nodal Agency under the Department of Space, IN-SPACE was created, which shall act as the promoter and regulator of space activities in India by non-government /private entities.
- It shall also allow utilization of capital intensive, DoS owned facilities at reasonable cost by the private sector.
- By 2022, the interim IN-SPACE Board has received close to 40 proposals from large industries, MSMEs, startups and academia, covering broad range of activities in space domain.

LIBERALISED DRONE RULES, 2021

- The Liberalized Drone Rules, 2021 were notified on 25th August, 2021.
- Several approvals were abolished, with total forms to be filled reduced from 25 to 5.
- Types of fees reduced from 72 to 4. For instance, the fee for remote pilot license has been reduced from Rs. 3,000 (for large drone) to Rs. 100 for all categories of drones.
- Drones up to 500 kg are now subject to regulations, compared to the earlier limit of 300 kg. This brings drone taxis and heavy payload carrying drones within the ambit of the drone rules.

- For drones with weight more than 500 kg, the provisions of the Aircraft Rules, 1937 will apply.
- A Digital Sky Platform has been developed as a single window platform for one step and one-time clearances for drone ownership and operation.



(TO BE CONTINUED...) 

(... CONTINUED FROM PREVIOUS SLIDE)

- Manufacturers and importers may generate their drone's Unique Identification Number on the Digital Sky Platform through self - certification.
- No remote pilot licences required for micro drones (for non-commercial use) and nano drones.
- Remote pilot licence will be issued by DGCA within 15 days of pilot receiving the remote pilot certificate from the authorized drone school through the Digital Sky Platform.
- Foreign-Owned and Controlled Indian Companies("FOCCs") can conduct drone operations in India.

- An interactive map on the Digital Sky Platform specifies color coded zones on the map, i.e., green, yellow and red, indicating - free zones, those which require prior permission and no fly zones respectively.
- The perimeters of these zones have also been liberalized to increase freely accessible airspace under the green category.
- The maximum penalty under new rules has been reduced from Rs. 5 lakhs to Rs. 1 lakh.



GEOSPATIAL DATA GUIDELINES WERE HIGHLY LIBERALISED!

- On 15th February, 2021, the Department of Science and Technology released guidelines for the creation, acquisition and use of geospatial data, including maps.
- Geospatial data is critical to the mapping activities. It was previously heavily regulated and required licenses to be obtained for the use of such data.



- Now several changes have been made and most important ones are .
 - ✓ All entities are now required to follow self certification process to show adherence to the guidelines as opposed to obtaining prior approval or licences for the use of geospatial data and maps.
 - ✓ Mapping activities are prohibited only for specific attributes of highly sensitive locations.



SVAMITVA IS GIVING e-PROPERTY CARDS IN RURAL AREAS!

SVAMITVA SCHEME

- It is Survey of Villages Abadi and Mapping with Improvised Technology in Village Areas (SVAMITVA).
- It was piloted in 2020, and now expanded to cover the entire Country.
- It maps rural residential and land ownership.
- It is mainly aimed to create non-disputable records.
- It leverages advanced technologies, such as Drones for measurements.
- It will be implemented in a phased manner over a period of 2020-24, and ultimately cover around 6.62 Lakh villages of the Country.

WHAT ARE THE BENEFITS?

- It provides farmers with non-disputable land records of land ownership.
- Hence, getting institutional credit through banks becomes easy.
- It also provides inherent security of having records of ownership.
- It reduces land disputes and litigation.

The Ministry of Panchayati Raj (MoPR) is the Nodal Ministry for implementation of the scheme.



A BRIEF ABOUT THE CATEGORISATION OF WATER AVAILABILITY

- 1** Central Ground Water Board (CGWB) is the apex organization dealing with ground water and related issues.
- 2** ➤ The main activities of CGWB are :-
 - ✓ Ground Water Management Studies
 - ✓ Ground Water Exploration Aided by Drilling
 - ✓ Monitoring of Ground Water Observation Wells.
- 3** The ground water resources are assessed in units i.e., blocks / talukas / mandals / watersheds.
- 4** ➤ There are four categories, namely -
 - ✓ “Safe” areas - which have ground water potential for development.
 - ✓ “Semi-critical” areas - where cautious groundwater development is recommended.
 - ✓ “Critical” and “Over-exploited” areas - where there should be intensive monitoring and evaluation and future ground water development be linked with water conservation measures.



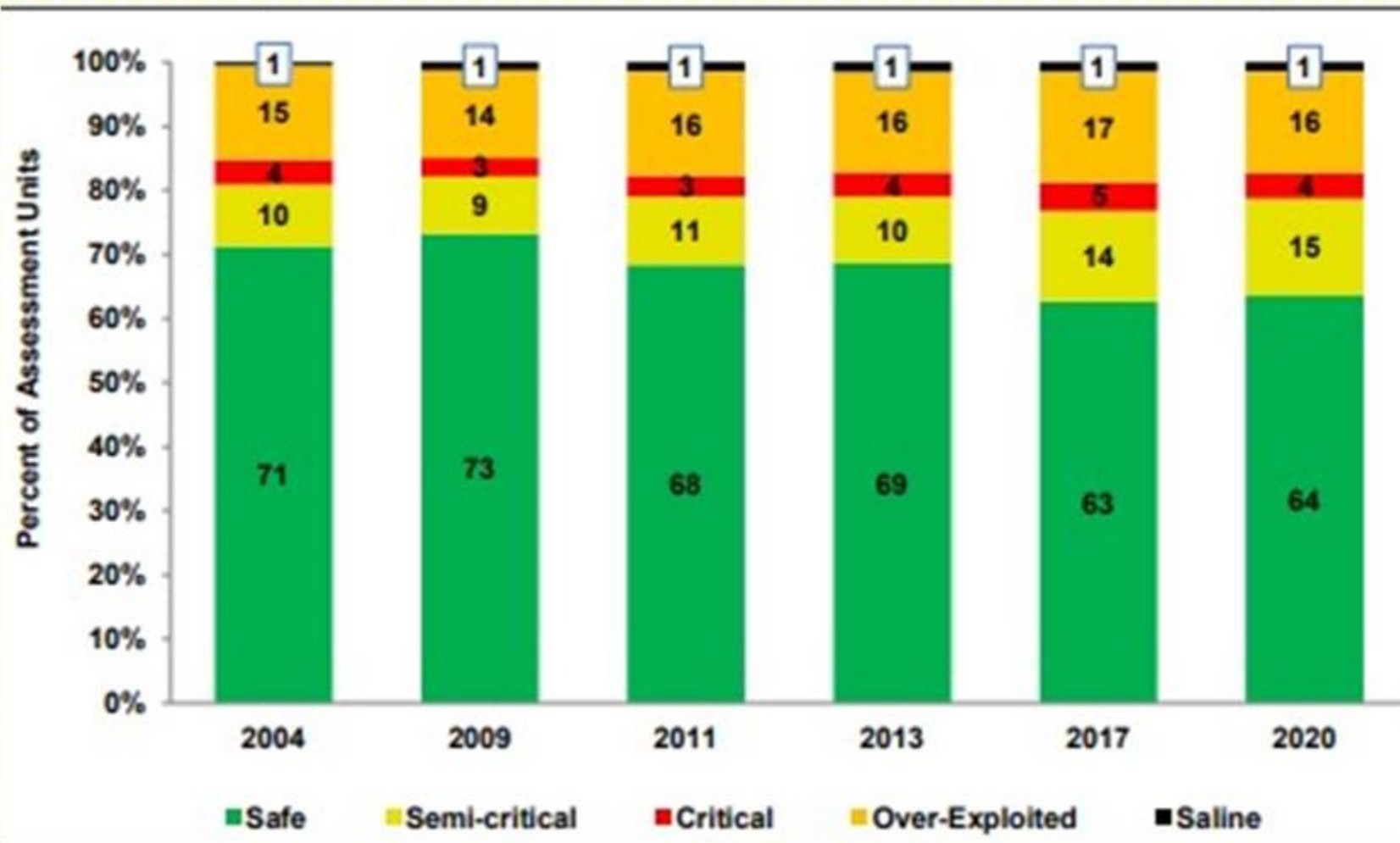
Central Ground Water Board (CGWB)

Ministry of Jal Shakti, Department of Water Resources, River Development and Ganga Rejuvenation

Government of India



A GLANCE AT GROUND WATER RESOURCE AVAILABILITY IN 2020



FOUR CATEGORIES

- Safe, if Stage of Extraction (SoE) less than 70%
- Semi critical, if SoE is greater than 70% and less than or equal to 90%.
- Critical, if SoE is greater than 90% and less than or equal to 100%.
- Over exploited, if SoE greater than 100%.

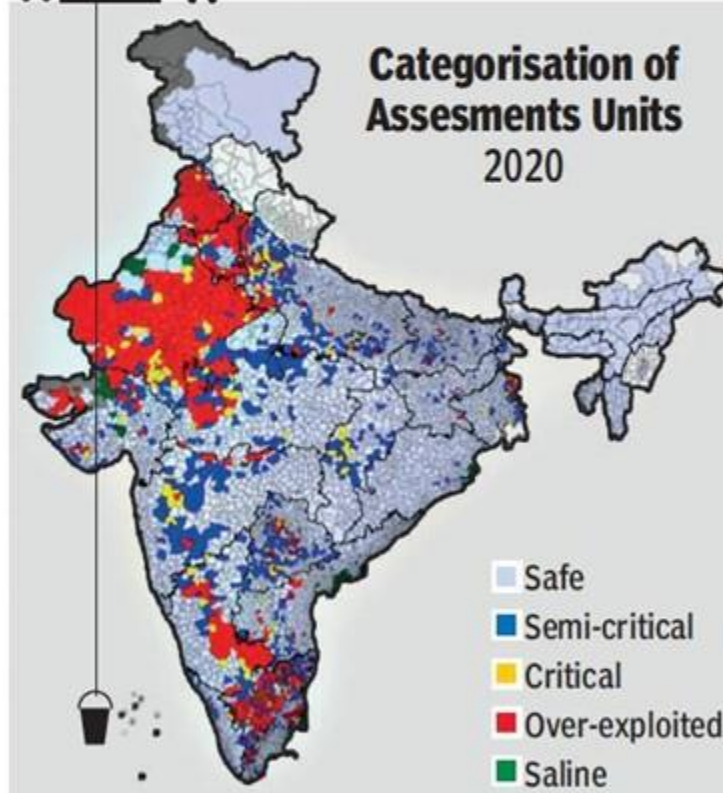
Ground water resources assessment of States/UTs is carried out jointly by State ground water/nodal departments and Central Ground Water Board (CGWB) at periodic intervals and the dynamic groundwater resources of India is published by compiling this data.



GROUND WATER SITUATION



Total no. of assessment units
2017: 6,881
2020: 6,965



2017 vs 2020

	No. of units	% of total
Safe		
2017	4,310	62.6
2020	4,427	63.6
Semi-critical		
2017	972	14.1
2020	1,057	15.2
Critical		
2017	313	4.6
2020	270	3.9
Over-exploited		
2017	1,186	17.2
2020	1,114	16
Saline		
2017	100	1.5
2020	97	1.3

*units: block/taluk/tehsil/mandal/firka; Source: Central Ground Water Board

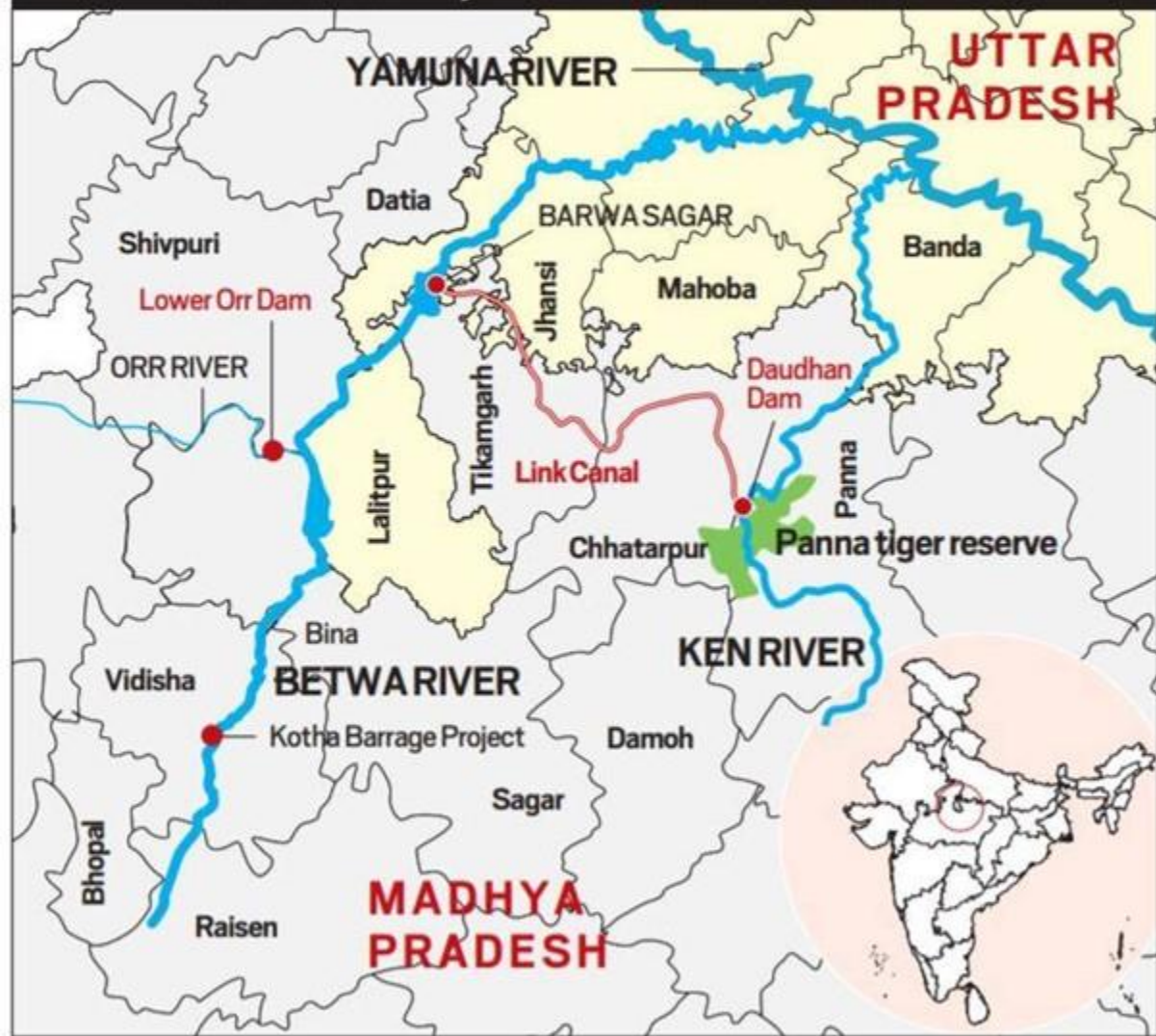
WATER USAGE IS MUCH LESS DISCUSSED IN INDIA!

- Out of the total 761 billion cubic meters (bcm) withdrawals of ground water, 688 bcm (close to 90%) goes to agriculture.
- China uses 385 bcm (64%) for agriculture out of the total withdrawals of 598 bcm.
- China's per unit land productivity in terms of crop production is almost 2 to 3 times that of India.



KEN-BETWA ... FIRST PROJECT UNDER NATIONAL PERSPECTIVE PLAN (NPP)!

TWO STATES, TWO RIVERS AND A LINK



KEN-BETWA RIVER LINKING PROJECT FINALLY TAKES OFF AND EXPECTED TO BENEFIT BUNDELKHAND REGION!



- Diversion of surplus waters of Ken basin to water deficit Betwa basin is planned.
- It involves deforesting a portion of Panna Tiger Reserve in Madhya Pradesh.

MEMORANDUM OF AGREEMENT WAS SIGNED ON MARCH 22, 2021

- On the World Water Day, on March 22, 2021, Memorandum of Agreement was signed among the Ministry of Jal Shakti and the Governments of Madhya Pradesh and Uttar Pradesh to implement Ken-Betwa Link Project (KBLP).
- It is the first project under the National Perspective Plan for interlinking of rivers.
- It envisages transferring water from the Ken river to the Betwa river, both the tributaries of Yamuna.

WHAT ARE THE IMPORTANT COMPONENTS OF KEN-BETWA LINK?

- Ken-Betwa link canal will be 221 km long.
- Phase-I involves Daudhan dam complex, and its subsidiary units.
- Phase-II involves
 - ✓ Lower Orr dam
 - ✓ Bina complex project
 - ✓ Kotha barrage
- The project is expected to provide annual irrigation of 10.62 lakh hectares, supply drinking water to 62 lakh people, and generate 103 MW of hydro power.
- The cost of the project is estimated at Rs. 35,111 Cr at 2017-18 prices.



TWO MORE IMPORTANT PROJECTS UNDER NPP ARE ON THE ANVIL!

DAMANGANGA-PINJAL



Damanganga - Pinjal river inter-linking project will take care of long-term water requirements of Mumbai. The project involves Gujarat and Maharashtra.

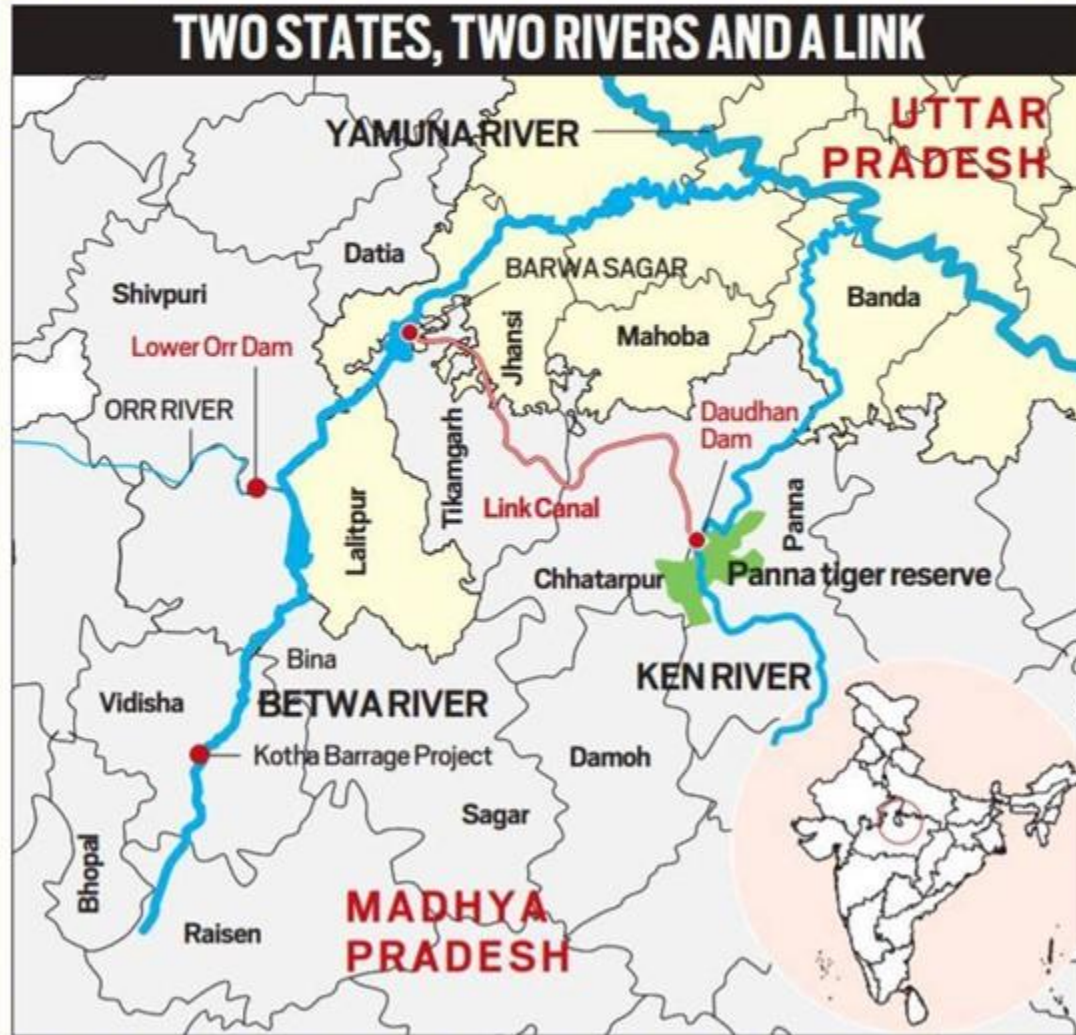
PAR-TAPI - NARMADA



Par-Tapi-Narmada Link proposes to transfer water from the water surplus regions of Western Ghats to the water deficit regions of Saurashtra and Kutch. The link project includes seven reservoirs proposed in north Maharashtra and south Gujarat.



THE ORIGIN OF NPP GOES BACK TO 1980!



LET US PERUSE THE PERSPECTIVE PLAN OF 1980!

- In 1980, Ministry of Irrigation prepared a **National Perspective Plan (NPP)** for water resources development.
- It envisaged **inter-basin** water transfer.
- It comprised two components - one is **Himalayan rivers** development and the other one is **Peninsular river** development.
- Based on this NPP, National Water Development Agency identified **30 river links** - **16** under the **Peninsular** component and **14** under the **Himalayan** component.

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For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

For Information:

info.learningspacedigital@gmail.com

Visit us at:

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98499 42299